Exam Date & Time: 02-Dec-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOV 2024

DIGITAL CIRCUITS [ECE 2127]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1) Implement a sequence detector using a Programmable Logic Array (PLA) and D flip-flops, based on the following equations: $D_1 = y_2x' + y_1y_2'x$, $D_2 = x$, and the output $Z = x'y_1y_2$. Here, y_1 and y_2 represent the outputs of the flip-flops. (4)

A)

B) Explain the process of converting a T flip-flop to a D flip-flop. (3)

C) Design and implement a 2-input XNOR gate using the minimum number of 2-input NOR gates and demonstrate the step-by-step process. (3)

2) Describe the operation of a full subtractor, including its logic and functionality. (4)

A)

B) Determine the logic expression implemented by the circuit shown in Fig. 2B. Simplify the expression to its reduced sum-of-products (SOP) form and implement the simplified expression using basic gates. (3)

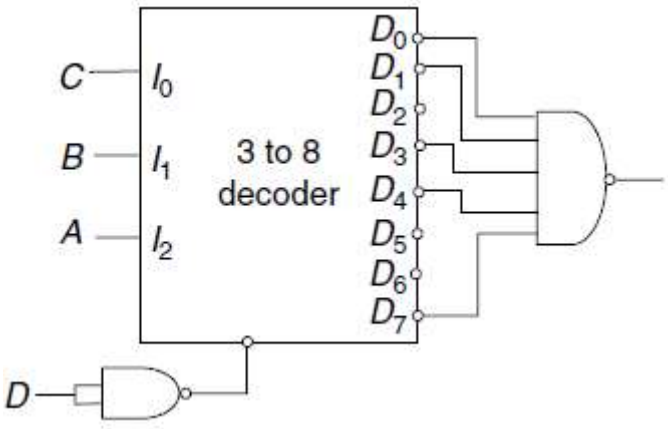


Fig. 2B.

C) Design a 3-bit odd parity generator and implement it using a 4-to-1 multiplexer. (3)

3) Design a 4-bit carry look-ahead adder using basic logic gates and explain each step of the design process. (4)

A)

B) Describe the operation of a positive edge-triggered D flip-flop constructed using 6 NAND gates, including its functional principles and behavior during clock transitions. (3)

C) Describe the operation of a 4-bit Ring counter utilizing positive edge triggered D flip-flops, and illustrate your explanation with a clear diagram and a sequence table. (3)

4) Design and implement a mod-7 up asynchronous counter using negative edge triggered T flip-flops and the necessary logic gates. Additionally, list the limitations associated with asynchronous counters. (4)

A)

B) Determine the output logic expression for the circuit depicted in Fig. 4B and simplify it to find the minimum sum-of-products (SOP) expression. (3)

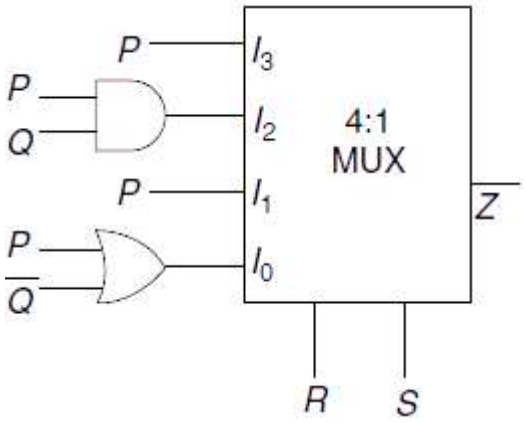


Fig. 4B

C) Design a synchronous Mod-6 up counter using positive edge triggered D flip flop and evaluate whether the counter is self-starting. (3)

5) Describe the operation of a 4-bit bi-directional shift register, detailing how data is shifted in both directions. (4)

A)

B) Design and implement a Mealy model overlapping sequence detector for the sequence 1010, using D flip-flops and any other necessary logic gates. (3)

C) For the state machine given in TABLE 5C, discover the equivalence partitions and write corresponding reduced state machine. PS, NS, X, and Z represent present state, next state, input, and output respectively. (3)

TABLE 5C

PS	NS, Z	
	X = 0	X = 1
A	E, 0	D, 1
B	F, 0	D, 0
C	E, 0	B, 1
D	F, 0	B, 0
E	C, 0	F, 1
F	B, 0	C, 0

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